

GU/CSE241/A1

Exercise Solutions

7.1 - Exercise Set (20 minutes)

```
(kali@DESKTOP-K00BQCF)-[/mnt/.../sem5/CSE241 - Security of Information Systems/assignments/1]
$ micro hash.cpp

(kali@DESKTOP-K00BQCF)-[/mnt/.../sem5/CSE241 - Security of Information Systems/assignments/1]
$ g++ ./hash.cpp -o hash.cpp.bin -lssl -lcrypto -lfmt && ./hash.cpp.bin file
SHA256(hash=9223904646ad6fb4fd4868ea25012003b800f1225704c7755fb6c7635ddb58fd, file=file)

(kali@DESKTOP-K00BQCF)-[/mnt/.../sem5/CSE241 - Security of Information Systems/assignments/1]
$ ./hash.cpp.bin file && python ./hash.py file && sha256sum ./file
SHA256(hash=9223904646ad6fb4fd4868ea25012003b800f1225704c7755fb6c7635ddb58fd, file=file)
SHA256(hash=684f6c373837ce4ff66dd877e9ec503ce3a7deea90a71eb48c9636894d67c491, file=./hash.py)
684f6c373837ce4ff66dd877e9ec503ce3a7deea90a71eb48c9636894d67c491 ./file

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$ micro hash.py

(kali@DESKTOP-K00BQCF)-[/mnt/.../sem5/CSE241 - Security of Information Systems/assignments/1]
$ ./hash.cpp.bin file && python ./hash.py file && sha256sum ./file
SHA256(hash=9223904646ad6fb4fd4868ea25012003b800f1225704c7755fb6c7635ddb58fd, file=file)
SHA256(hash=684f6c373837ce4ff66dd877e9ec503ce3a7deea90a71eb48c9636894d67c491, file=file)
684f6c373837ce4ff66dd877e9ec503ce3a7deea90a71eb48c9636894d67c491 ./file

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$ g++ ./hash.cpp -o hash.cpp.bin -lssl -lcrypto -lfmt && ./hash.cpp.bin file
SHA256(hash=684f6c373837ce4ff66dd877e9ec503ce3a7deea90a71eb48c9636894d67c491, file=file)

(kali@DESKTOP-K00BQCF)-[/mnt/.../sem5/CSE241 - Security of Information Systems/assignments/1]
$ ./hash.cpp.bin file && python ./hash.py file && sha256sum ./file
SHA256(hash=684f6c373837ce4ff66dd877e9ec503ce3a7deea90a71eb48c9636894d67c491, file=file)
SHA256(hash=684f6c373837ce4ff66dd877e9ec503ce3a7deea90a71eb48c9636894d67c491, file=file)
684f6c373837ce4ff66dd877e9ec503ce3a7deea90a71eb48c9636894d67c491 ./file

(kali@DESKTOP-K00BQCF)-[/mnt/.../sem5/CSE241 - Security of Information Systems/assignments/1]
$ #done
```

```
(kali@DESKTOP-K00BQCF)-[/mnt/.../CSE241 - Security of Information Systems/assignments/1/P1]
$ ./hash.cpp.bin content && python ./hash.py content && sha256sum ./content
SHA256(hash=f20b58ecdd832a59d9a1687d0f2b14ed72e0c21bb5467d57cc7a0703add3699f, file=content)
SHA256(hash=f20b58ecdd832a59d9a1687d0f2b14ed72e0c21bb5467d57cc7a0703add3699f, file=content)
f20b58ecdd832a59d9a1687d0f2b14ed72e0c21bb5467d57cc7a0703add3699f ./content

(kali@DESKTOP-K00BQCF)-[/mnt/.../CSE241 - Security of Information Systems/assignments/1/P1]
$ micro content

(kali@DESKTOP-K00BQCF)-[/mnt/.../CSE241 - Security of Information Systems/assignments/1/P1]
$ hexedit content

(kali@DESKTOP-K00BQCF)-[/mnt/.../CSE241 - Security of Information Systems/assignments/1/P1]
$ ./hash.cpp.bin content && python ./hash.py content && sha256sum ./content
SHA256(hash=99df935ed5665cb919b7e548fb49abef3867c881e6d7a5e5384b8342bb7b25fc, file=content)
SHA256(hash=99df935ed5665cb919b7e548fb49abef3867c881e6d7a5e5384b8342bb7b25fc, file=content)
99df935ed5665cb919b7e548fb49abef3867c881e6d7a5e5384b8342bb7b25fc ./content

(kali@DESKTOP-K00BQCF)-[/mnt/.../CSE241 - Security of Information Systems/assignments/1/P1]
$ micro ../P2/diff

(kali@DESKTOP-K00BQCF)-[/mnt/.../CSE241 - Security of Information Systems/assignments/1/P1]
$ cd ../P2

(kali@DESKTOP-K00BQCF)-[/mnt/.../CSE241 - Security of Information Systems/assignments/1/P2]
$ ls
total 4.0K
-rwxrwxrwx 1 kali kali 854 Oct 29 18:41 calc_diff_bits.sh
-rwxrwxrwx 1 kali kali 130 Oct 29 18:46 diff

(kali@DESKTOP-K00BQCF)-[/mnt/.../CSE241 - Security of Information Systems/assignments/1/P2]
$ cat diff
f20b58ecdd832a59d9a1687d0f2b14ed72e0c21bb5467d57cc7a0703add3699f
99df935ed5665cb919b7e548fb49abef3867c881e6d7a5e5384b8342bb7b25fc

(kali@DESKTOP-K00BQCF)-[/mnt/.../CSE241 - Security of Information Systems/assignments/1/P2]
$ ./calc_diff_bits.sh $(cat ./diff | head -n1) $(cat ./diff | tail -n1)
Bits changed: 152/512

(kali@DESKTOP-K00BQCF)-[/mnt/.../CSE241 - Security of Information Systems/assignments/1/P2]
$ #done
```

```

real    0m3.813s
user    0m0.674s
sys     0m0.149s
DONE.. I suppose you now see the diff, man.

(kali@DESKTOP-K00BQCF) - [/mnt/.../assignments/1/P1/P3]
$ micro sha_diff_speeds.sh

(kali@DESKTOP-K00BQCF) - [/mnt/.../assignments/1/P1/P3]
$ ./sha_diff_speeds.sh ./a_500MB_file
LoAdInG file ./a_500MB_file...
Start Calculating Performance...
Calculate[SHA256]
cl9bafdc2e8bf59851c4b98c3fc40bcc390b25541f01195c0a7c35e7887ebf46a ./a_500MB_file

real    0m3.879s
user    0m0.936s
sys     0m0.190s
\n
Calculate[SHA512]
^C

(kali@DESKTOP-K00BQCF) - [/mnt/.../assignments/1/P1/P3]
$

(kali@DESKTOP-K00BQCF) - [/mnt/.../assignments/1/P1/P3]
$ micro sha_diff_speeds.sh

(kali@DESKTOP-K00BQCF) - [/mnt/.../assignments/1/P1/P3]
$ ./sha_diff_speeds.sh ./a_500MB_file
LoAdInG file ./a_500MB_file...
Start Calculating Performance...
Calculate[SHA256]
9bafdc2e8bf59851c4b98c3fc40bcc390b25541f01195c0a7c35e7887ebf46a ./a_500MB_file

real    0m3.921s
user    0m0.969s
sys     0m0.153s

Calculate[SHA512]

```



Exercise 4: SHattered Attack Research

- SHattered is a Cryptanalyst for SHA1 that produced a two different pdfs witht he SAME SHA1-hash!
- This is becasue collisions in SHA1, you can see the full paper here: shattered.io

7.2 - Quiz Solutions

1. Three Main Security Properties of Hash Functions

1. **Pre-image Resistance:** Given a hash value h , it should be computationally infeasible to find any message m such that $h = \text{hash}(m)$
2. **Second Pre-image Resistance:** Given an input m_1 , it should be computationally infeasible to find a different input m_2 such that $\text{hash}(m_1) = \text{hash}(m_2)$
3. **Collision Resistance:** It should be computationally infeasible to find two different messages m_1 and m_2 such that $\text{hash}(m_1) = \text{hash}(m_2)$

2. Why SHA-1 is No Longer Recommended

- As i mentioned in Ex.4, google produced two diff. pdfs with the same sha1-hash, so it's broken, or has a problem in collision resisten.
- Also, Theoretical vulnerabilities discovered as early as 2005
- NIST deprecated SHA-1 in 2011 and disallowed its use after 2013 # that's logicaly the normal flow after shattered.io papers

3. Why SHA-256 Can't Be Used Directly for Password Storage

- I see that becasue the powerfull of todays super computers, we can make brute-force attacks on it.
- Also the SHA256 is fast to compute, so the brute force will be practical
- Also i we want to use it, we have to salt it for preventing rainbow attacks

4. Avalanche Effect

The avalanche effect is a property where a small change in input (even one bit) causes a significant change in the output hash (approximately 50% of bits change). This ensures that similar inputs produce completely different hashes.

5. Difference Between SHA-256 and HMAC-SHA256

- **SHA-256:** A cryptographic hash function that takes input and produces a fixed-size output
- **HMAC-SHA256:** A keyed-hash message authentication code that uses SHA-256 as the underlying hash function but includes a secret key for authentication

Additional Resources

- `std::fmt`
- Python hashlib Documentation
- SHAttered Attack Website

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